

Predicting Contract Renewal in Public Digital Procurement

A Survival-Analysis Approach on BOAMP Notices for Gigalis
Internship Report — Quick Orientation to What Was Done and Where

Internship Report — Gigalis (Pays de la Loire)

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Abstract

This report is the **quick-reading guide** to the internship: it is meant to let a reader scan the project rapidly and understand *what was done*, *why it matters*, and *where each workstream sits* in the overall pipeline. The detailed implementation, audit logic, intermediate diagnostics, and methodological justifications are intentionally left to the separate **technical report** (`phase1_technical_report.pdf`).

The internship work concerns predictive analysis of public digital procurement for **Gigalis**, a central purchasing body in Pays de la Loire. The goal is to move from a descriptive view (“what happened?”) towards a predictive one (“when will a contract be re-tendered?”). Because BOAMP contains *no explicit renewal field*, the renewal relationship must be reconstructed from observable signals. We build a proxy renewal event using Sentence-Transformer text embeddings combined with CPV, temporal, and buyer signals, and use it as the event variable for survival analysis. On 1,100 eligible source contracts the linking method recovers a plausible renewal for 705 of them (**64.1%**), against an 11.3% lexical baseline. Survival modeling gives a global Kaplan–Meier median survival of **48.2 months**, a Cox concordance of **0.6541**, and a best-fitting log-normal parametric model. The fitted model is then translated into 12- and 24-month renewal-risk indicators at contract, buyer, and segment level.

1 How to Read This Report

This document is a **high-level navigation guide**. Each section states *what* was done, *what the key result is*, and *where* the outputs live. When you need the exact field logic, preprocessing rules, linking formula, diagnostics, or full model specification, go directly to the **technical report**.

Pipeline at a glance

Table 1: Component status and key outputs. Status: **Done**, **Partial**, **Not started**.

Component	Status	Key result	Where
Data cleaning & EDA	Done	3,181 notices → 1,933 source contracts	data/processed/boamp_full_clean.csv
Renewal linking (event reconstruction)	Done	705/1,100 eligible linked (64.1%)	boamp_renewal_linking_quality/
Survival dataset construction	Done	1,100 rows × 36 cols	data/processed/boamp_phase2_survival.csv
Kaplan–Meier & log-rank	Done	Median 48.2 mo; duration significant	notebooks/02_survival_modeling_boamp.ipynb
Cox PH model	Done	C-index 0.6541	same notebook + reports/tables/survival/
Parametric AFT models	Done	Log-normal best (AIC 7,176.5)	same notebook
Sensitivity analysis	Done	Ranking stable across 3 definitions	reports/tables/survival/sensitivity_comparison.csv
Robustness checks	Done	Placebo, bias check, threshold sweep	validation_robustness/
Operational risk indicators	Done	12/24-mo risk at contract/buyer/segment	reports/tables/survival/top20_renewal_risk.csv
Technology taxonomy	Done	Rule-based (CPV + keywords), 11 categories	data/processed/taxonomy.csv
Manual event validation	Partial	Sample drawn; review not done	event_validation/
Trained NLP classifier	Not done	Taxonomy is rule-based only	—
Change-point detection (Phase 4)	Not done	Scoped but not implemented	—

2 Context and Problem

2.1 Why this project

Gigalis operates a central purchasing body for digital services in Pays de la Loire. To anticipate re-tendering workloads, it wants to predict *when* digital contracts come up for renewal. The core obstacle: BOAMP publishes calls for tenders and awards, but *no notice says “this renews contract X”*. The renewal link must be inferred from text, classification, timing, and buyer identity.

2.2 Scope

- **Sectors:** CPV divisions 48 (software), 72 (IT services), 32 (telecom), 35 (security).
- **Territory:** departments 44, 49, 53, 72, 85 (Pays de la Loire).
- **Period:** 2015–2024. Study end date: 2024-12-31.
- **Modeling path:** BOAMP only. DECP is documented but not linked (no shared key between the two sources).

3 Data

3.1 Corpus numbers

- 3,181 BOAMP notices downloaded (2015–2024).
- 1,933 are APPEL_OFFRE (source contracts); 1,081 are ATTRIBUTION.
- 1,100 are *eligible*: their expected renewal window falls before 2024-12-31.
- 833 are structurally censored upfront (too recent to observe a renewal).

DECP (3,039 digital contracts, same scope) is far richer (100% SIRET, 99.4% durations) but cannot be merged to BOAMP at contract level. Contract-level DECP merging remains a future path. Buyer-name SIREN enrichment (API Recherche d’Entreprises) was completed separately on 2026-06-24 and is described in the next-steps section below.

3.2 Key data quality issues (modeling consequences)

Table 2: Key BOAMP data-quality findings and their modeling consequence.

Finding	Value	Consequence
Buyer SIRET coverage	~9% of corpus	Grouping uses normalised names; same entity may fragment into several keys.
Amount fill rate (APPEL_OFFRE)	~21%	Amount excluded from linking signals.
Declared duration granularity	69% at 12/24/36/48 mo	Duration used as a covariate, not as survival target.
Imputed durations	25% of eligible cohort	Flagged with <code>dur_was_imputed</code> ; imputation is 48 months.
No explicit renewal label	structural	<code>event</code> is a proxy reconstructed by the linking algorithm.

Full field-level cleaning rules and missingness tables are in the technical report (§2) and `reports/data_quality_report/`.

3.3 Taxonomy

Each contract is tagged with one of eleven technology categories (Software & Applications, IT Services & Consulting, Telecom & Networks, Cybersecurity, Cloud & Infrastructure, Data & AI, etc.) via a **deterministic rule-based mapping** from CPV codes and description keywords. This is *not* a trained classifier — the planned NLP classifier was not implemented. Output: `data/processed/taxonomy.csv`.

4 What Was Built

4.1 Phase 1 — Renewal-event reconstruction

Each eligible source contract is matched against later BOAMP notices from the same buyer using a four-signal composite score: semantic text embeddings (Sentence-Transformer, model `paraphrase-multilingual-MiniLM-L12-v2`), CPV hierarchy match, temporal proximity, and buyer-identity agreement (weights 0.40 / 0.25 / 0.20 / 0.15). The best candidate above the hard filter threshold is selected as the renewal successor. See technical report §3.2 for the full formula and weight derivation.

Table 3: Renewal-linking outcome.

Quantity	Value	
Source contracts (APPEL_OFFRE)	1,933	
Eligible (renewal window observable)	1,100	
Linked renewals (<code>event = 1</code>)	705	(64.1%)
Right-censored (<code>event = 0</code>)	395	(36.6%)
Lexical baseline linking rate	11.3%	

Linking rate by category (Table 4): Software & Applications and IT Services & Consulting renew most (71%, 68%); Data & AI and IT Hardware least (41%, 48%). Small categories ($n < 10$) are underpowered.

Table 4: Event rate by technology category (1,100 eligible contracts).

Category	n	events	rate %
IT Services & Consulting	312	211	67.6
Software & Applications	209	148	70.8
Telecom & Networks	204	120	58.8
Cybersecurity	121	76	62.8
Unknown	108	58	53.7
Digital Workplace & Collaboration	45	29	64.4
Data & AI	34	14	41.2
IT Maintenance & Support	22	14	63.6
IT Hardware & Equipment	21	10	47.6
Cloud & Infrastructure	19	13	68.4
GIS & Mapping	5	4	80.0

Link quality. Of the 705 accepted links: 133 are single-candidate (no competing match); of the remaining 572 multi-candidate links, 265 (46%) have a score margin below 0.05, flagging an ambiguous winner. Of the 705 links, 100 (14.2%) satisfy the `high_confidence_strict` criterion ($S \geq 0.70$ and margin ≥ 0.05 , multi-candidate only); these form the Model C core (Table 7). This low high-confidence rate motivates the manual validation step (not yet done; see Section 5).

Main unlinked failure mode. 339 of 395 unlinked contracts (84%) have no temporal partner in the search window — not a scoring failure, but a true absence of a candidate notice in the observable period.

Outputs: `boamp_renewal_linking_quality/outputs/` (`boamp_renewal_links.csv`, `boamp_renewal_candidates.csv`, `boamp_linking_stats.csv`).

4.2 Phase 2 — Survival analysis

The survival dataset (`data/processed/boamp_phase2_survival.csv`) has one row per eligible contract with observed/censored duration and covariates (`declared_duration_months`, `dur_was_imputed`, `start_year`, category, procedure type). Three model families were fitted in `notebooks/02_survival_modeling_boamp.ipynb`.

Kaplan–Meier. Global median survival: **48.2 months**, consistent with the prevalence of 48-month framework contracts. Log-rank tests (Table 5): renewal timing is driven by declared duration and imputation status ($p < 0.001$), not by technology category ($p = 0.15$).

Table 5: Log-rank tests for differences in renewal timing.

Stratifying variable	statistic	p	df
Category (multivariate)	6.75	0.150	4
Duration was imputed	24.41	< 0.001	1
Declared-duration group	62.43	< 0.001	2
Procedure type (collapsed)	2.10	0.349	2

Cox proportional-hazards model. In-sample concordance: **0.6541**. Two predictors are significant (Table 6): longer declared duration lowers hazard (HR 0.991 per month), and later start year raises it (HR 1.094 per year). No category or procedure effect.

Table 6: Cox model: significant covariates (multivariate).

Covariate	HR	95% CI	p
<code>declared_duration_months</code>	0.991	[0.987, 0.995]	< 0.001
<code>start_year</code>	1.094	[1.050, 1.140]	< 0.001
<code>dur_was_imputed</code>	0.864	[0.718, 1.040]	0.123
Category / procedure terms	not significant ($p > 0.1$)		

The positive `start_year` HR is partly structural: 2021–2024 contracts are right-censored because a typical 48-month contract would renew after the 2024-12-31 study close.

Parametric AFT models. Log-normal fits best (AIC 7,176.5, concordance 0.672), outperforming log-logistic (7,248.8) and Weibull (7,380.3). Full parametric results: `reports/tables/survival/parametric_aic_comparison.csv`.

Sensitivity analysis. Three event definitions were tested (Table 7). The Cox concordance is stable across all three, confirming that the *ranking* of renewal risk is robust even under the strictest definition.

Table 7: Sensitivity to event definition. “—” = KM median never reached. KM median in months.

Variant	events	rate %	KM median	Cox C
A — baseline (all links)	705	64.1	48.2	0.6541
B — conservative (≥ 0.50)	497	45.2	—	0.6061
C — strict H.C. ($S \geq 0.70$, $\Delta \geq 0.05$, multi-cand.)	100	9.1	—	—

Full results: `reports/tables/survival/` (9 CSV files: univariate, multivariate, log-rank, AIC, sensitivity, top-20 risk ranking).

Proportional hazards assumption. Schoenfeld test flags a violation for `declared_duration_months` ($p < 10^{-5}$); all other covariates pass. Stratification or a time interaction for duration is flagged as future work. The log-normal AFT avoids the PH assumption and gives the same conclusions.

4.3 Robustness and validation checks

These are in `validation_robustness/` (untracked — results exist but not merged to main git history yet).

- **Threshold sensitivity (7 scenarios, score 0.35 to 0.90):** Cox concordance stays above 0.59 throughout; directional findings stable. Full table: `validation_robustness/outputs/threshold_sensitivity_summary.csv`.
- **Placebo check:** winning link mean score 0.579 vs. runner-up 0.435 vs. synthetic no-buyer 0.425 — scoring discriminates.
- **Event/censoring bias:** SMD = 0.13 (duration), -0.03 (start year) between event and censored groups — covariates balanced; dominant bias is structural right-censoring of 2022–2024 contracts.
- **Manual validation:** 150-row sample drawn (`event_validation/outputs/manual_validation_sample.csv`), protocol documented (`event_validation/event_validation_protocol.md`). **Record-level review not yet executed.** Precision/recall of the linking step therefore unknown.

4.4 Operational renewal-risk indicators

The log-normal AFT model is applied to all 1,097 scored contracts to produce conditional renewal probabilities at 12 and 24 months ahead.

Table 8: Operational risk summary (1,097 scored contracts).

Metric	Value
Mean 12-month renewal probability	0.190
Mean 24-month renewal probability	0.332
Max 12-month probability (any contract)	0.435
Contracts in <i>medium</i> band (0.25–0.50)	222
Contracts in <i>low</i> band (< 0.25)	875
Expected renewals within 12 months	208
Expected renewals within 24 months	365
Top buyer by expected 12-month volume	Pays de la Loire (18.6)
Top segment by expected 12-month volume	IT Services & Consulting (61.6)

These are prioritisation indicators, not legally verified renewal probabilities. Output: `reports/tables/survival/top20_renewal_risk.csv`.

5 What Was Not Done

- **Manual validation of linked pairs.** The 150-row sample is ready; the review has not been executed. Until done, there are no independently verified precision/recall estimates for the linking step.
- **Trained NLP classifier.** The technology taxonomy is rule-based (CPV + keywords). The planned learned classifier was not implemented.
- **Change-point / trend detection (Phase 4).** Scoped but not started. No code or outputs exist.
- **DECP semantic linking.** The Sentence-Transformer method was applied to BOAMP only. DECP linking remains at the 8.3% lexical prototype.
- **Live prediction scoring.** The current model evaluates contract age as of the study end date. A production run would need to re-evaluate each contract at a live prediction date.

6 Key Limitations

- **Proxy event.** `event = 1` means a plausible renewal link was found, not a legally certified renewal. Most links are low-to-medium confidence (62.4% low, 10.2% medium, 8.5% high in the four-tier system).
- **Weak buyer identification.** Only $\sim 13\%$ of buyers carry a SIRET; the rest are matched by normalised name. Spelling variants (“Ville de Nantes” vs “Nantes Métropole”) create false negatives. The 2024 eForms notices do include SIRET but are all censored upfront. After SIREN enrichment (completed 2026-06-24), 660 of the 1,100 phase-2 eligible rows carry a HIGH-confidence SIREN-enriched buyer identity in `boamp_phase2_survival_siren_enriched.csv`; 440 remain on the name-based key.
- **Short-text false positives.** Generic descriptions (“prestations informatiques”) can match spuriously.
- **Structural right-censoring.** Contracts from 2021–2024 are largely censored; the positive `start_year` hazard partly reflects study truncation, not a genuine cohort trend.
- **Modest discrimination.** Concordance ≈ 0.65 — useful for prioritisation, not sufficient for high-stakes individual decisions.

7 Conclusion and Next Steps

The internship delivered a reproducible end-to-end pipeline: BOAMP notice acquisition $\rightarrow$ cleaning $\rightarrow$ renewal-event reconstruction $\rightarrow$ survival modeling $\rightarrow$ operational risk indicators. The main methodological result is lifting the renewal linking rate from 11.3% (lexical baseline) to 64.1% using Sentence-Transformer embeddings. Survival modeling shows renewal timing is driven by declared contract duration (median 48.2 months) rather than technology family, and the log-normal AFT model translates this into actionable 12/24-month renewal-risk rankings.

Immediate next steps (prioritised):

1. **Execute manual validation** of the 150-row sample to obtain independently verified precision/recall estimates for the linking step (`event_validation/`).
2. **Re-score contracts at a live prediction date** (current scores use end-of-study ages).
3. **SIREN buyer enrichment** — *completed 2026-06-24*. 275 of 525 unique buyers (52.4%) matched with HIGH confidence via API Recherche d'Entreprises; 23 previously fragmented buyer groups merged. An **auxiliary Jaccard comparison** (separate from the official Sentence-Transformer phase-2 pipeline) evaluated the effect of SIREN resolution on the Jaccard method: the Jaccard event rate rose from 11.3% to 12.1% (+15 events; log-rank $p = 0.47$, not significant). The two Jaccard output files — `boamp_full_survival.csv` (name-based baseline, 1,933 rows) and `boamp_full_survival_enriched.csv` in `boamp_renewal_linking_siren_enriched/outputs/` — are **legacy comparison artifacts**, not the current-stage survival dataset.

The official phase-2 survival dataset (`data/processed/boamp_phase2_survival.csv`, 1,100 rows, 705 events) has been **annotated** with buyer identity enrichment metadata in `data/processed/boamp_phase2_survival_siren_enriched.csv`. This annotation adds four columns (`buyer_key_enriched`, `enriched_siren`, `enrichment_confidence`, `enrichment_score`) via left-join; it does **not** recompute the event variable. Enrichment-confidence breakdown over the 1,100 phase-2 rows: **HIGH 660 (60.0%)**, MEDIUM 77, LOW 45, NO_MATCH 318. The 440 unresolved rows remain on the original name-based buyer key. **Caution:** HIGH-confidence automated enrichment reflects strong textual similarity under the chosen normalisation scheme; it is *not equivalent to manual validation* and rare false positives at HIGH confidence remain possible. See `buyer_siren_enrichment/` and the Data Quality Report.

4. **Replace rule-based taxonomy** with a trained text classifier.
5. **Implement change-point detection** (Phase 4).